# ROBOTICS CLUB MNNIT
# ANNUAL SUMMER PROJECT
# VISION GUIDED ROVER
# (Phase – 1)

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## About the Project

The Vision Guided Rover is an autonomous navigation rover capable of navigating between a set of user defined points with help of Aruco markers placed in the environment. The entire mechanical structure was designed using **SolidWorks** and fabricated using **3D printing technology**, ensuring a lightweight yet durable frame.

The prime movers in this rover are 4 N25 metal gear encoder motors which are controlled by an Arduino Uno which acts as its microcontroller (MCU). Image processing techniques, PID control algorithms and differential drive mechanics are used to achieve accurate navigation, precise control, proper detection and odometry. The software of the rover is programmed in Python and C++ languages.

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## Team & Mentorship

The Vision Guided Rover project was developed by a dedicated team of **second-year students** under the guidance of senior mentors (from third year and final year) from the Robotics Club, MNNIT Allahabad.

### Mentors

- Sudhanshu Ranjan (ECE, Final Year)
- Dhruv Chandhok (MED, 3rd Year)
- Kushagra Verma (ECED, 3rd Year)
- Abhay Agarwal (MED, 3rd Year)

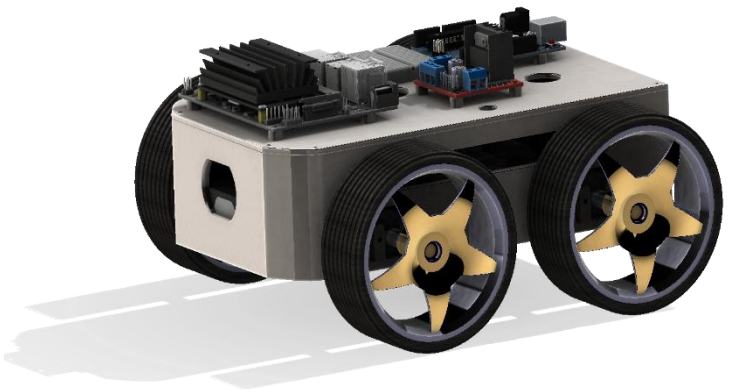
### Contributors

- Vedang Tripathi (MED, 2nd Year)
- Divyansh Pandey (EED, 2nd Year)
- Kritika Parihar (ECED, 2nd Year)
- Harsh Agarwal (CSED, 2nd Year)

- Kumaradithya (CSED, 2nd Year)
- Aman Shukla (MED, 2nd Year)

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## CAD Design



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## Tech Stack

- Arduino IDE (embedded control, odometry & motor actuation)
- Python-OpenCV (Image Processing and Pose Estimation)
- NumPy (Matrix calculations)
- PID Control (Control motors for stable and accurate motions)
- Ubuntu-Debian (On Board Processing and script execution)
- SolidWorks (mechanical design & CAD modeling)
- Serial Communication (Data transfer between Jetson and Arduino via USB)

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## Components Used

- Arduino Uno (Microcontroller)
- NVIDIA Jetson Nano 2GB (On board processing)
- Lenovo FHD USB Webcam
- N25 DC Metal Gear Encoder Motors
- Breadboard
- Lithium-ion battery (12.5V, 3000mAh)
- Jumper wires
- Wheels
- Nuts and bolts
- PLA filament (3D-printed chassis & joints)

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## Development

The development of this project begun with camera calibration and detection of aruco markers, identification and performing pose estimation.

On the other hand, the main body of the rover i.e.chassis that holds all the components was

designed through 3D CAD Modelling. Once the design was completed, the components were fabricated using 3D printing and motors and other components were assembled over it.

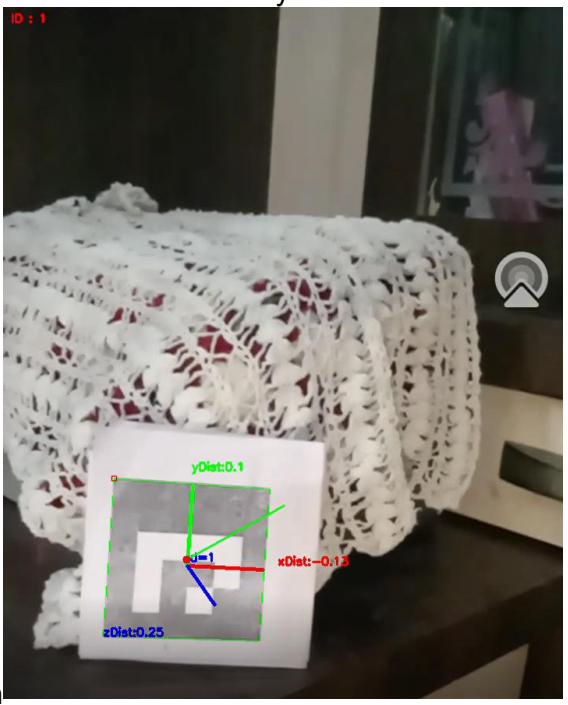
The odometry part of the rover is achieved with the help of encoders placed on each motor. Odometry refers to estimating the rover's position in terms of x–y coordinates and orientation angle with respect to a defined origin. The encoders track the angle by which the wheel is rotated. This information about each wheel is converted into the robot's position and orientation by geometrical operations under Differential Drive mechanics.

In the last the rover was fine tuned with a set of parameters over numerous testing and iterations.

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## Aruco Detection and Pose Estimation

ArUco markers are square shaped binary images used in computer vision for augmented reality, robotics, and pose estimation, featuring a wide black border and an inner binary matrix that encodes a unique ID, allowing for easy detection and robust tracking in images. These markers are composed of black (0) and white (1) squares, forming a grid (often 5x5 or 7x7) that contains data bits and parity bits for error correction, enabling precise identification and 3D positioning.




In pose estimation the position of the aruco marker is found with respect to the rover's camera. From which the distances in x, y and z directions, angle by which the marker is turned about the rover in degrees is calculated. The calculations are required for the rover to approach and align towards the marker.

The pose information of rover with respect to marker is sent through USB from Jetson to Arduino which then implements PID control over the motors to make the rover steer and cruise towards the marker.

The Jetson continuously sends the distance from the marker to the Arduino and once the rover is close enough to the marker it sends a signal to Arduino to make a turn and it control the motors accordingly and the rover turns left or right depending on its path.

## Steering and Turning

The steering of the rover is achieved by reducing and increasing the speeds of the left and right motors respectively for a left steer and vice versa for a right turn.

While the rover turns are exact 90 degree angles by reversing the motors direction on either side at same speed for certain duration with the help of encoders.

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## Flow of processes

1. The rover is given the Aruco marker IDs with their respective position coordinates and orientation angle about vertical in degrees.

2. If next aruco marker is visible the rover calculates the position, steers and approaches the marker with PID control. When it reaches close to it, it takes left/right turns according to the next markers coordinates.

3. If the marker is not visible, it used encoder data and utilizes odometry for moving towards the next marker's position coordinates. Once the marker gets detects is executes step 2.

4. It reaches each marker sequentially in this way. When it detects the last marker, the rover aligns itself about marker and docks in front of it. Now the navigation is done and the rover shuts down.

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## Key Highlights

- Smooth motion on both rough and smooth surfaces
- Custom-designed and 3D-printed chassis
- On board processing
- PID Control
- High torque stable motors

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## Real-World Relevance

Vision-guided autonomous rovers similar to this project are widely used in structured indoor and controlled environments where **marker-based localization and vision feedback** provide reliable navigation without heavy infrastructure:

- Industrial automation & smart factories
- Campus, hospital, and indoor delivery robots
- Robotics research & education platforms
- Surveillance and inspection tasks

This project closely mirrors the **core architecture used in real AGVs**, combining onboard vision processing (Jetson), low-level motor control (Arduino), and closed-loop feedback systems, making it a strong foundation for scalable autonomous robotic systems.

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## Challenges & Learnings

While developing the Vision Guided Rover, the team encountered several real-world engineering challenges that contributed significantly to the learning process. The major challenge was the serial communication between Jetson and Arduino. Due to high processing rate the communication often broke in between resulting the program to crash. This was fixed by limiting the processing rate, turning camera off when it was not needed, and limiting the frame rate.

The next major challenge face was involved in odometry. To estimate the position accurately several times testing was done. Fine tuning was done. The drifts from encoders caused errors which led to mild deviation from accuracy.

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## Future Goals in Phase-2

- Integrate Robot Operating System 2 (ROS2) for more robust operations and data transmission
- Add SLAM for localization and mapping
- Include obstacle detection and avoidance
- Add Nav2 for path planning
- Fuse IMU with encoders for more precise odometry

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## Github Repository:

Access the Github repo for this project either by scan the given QR or through the link provided below.

https://github.com/vedangtripathi45/Vision-Guided_Rover-2024-2025-Project

## Implementation Poster:

# MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY ALLAHABAD

## VISION GUIDED ROVER

**SUMMER PROJECT 2025**

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**Mentors:** Dhruv Chandhok, Kushagra Verma, Shaan Kapoor

### OBJECTIVE

To design an autonomus navigation bot that traverses following the Aruco placed from the starting to the docking end.

### DESCRIPTION

The Vision Guided Rover is a low-cost, autonomous navigation solution leveraging ArUco marker detection and PID-based motor control. It detects visual markers, estimates relative position and orientation, and adjusts its path accordingly to move and dock without human intervention. The project integrates computer vision and embedded control using OpenCV, Python, Embedded C, and Jetson Nano.

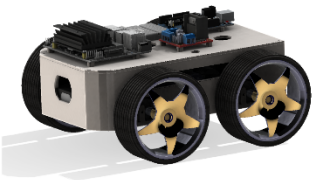


Figure: CAD Model of Vision Guided Rover

### SOFTWARE DEVELOPMENT

Implemented Computer Vision using OpenCV to detect Aruco Marker with its ID and configure its estimated pose to determine its distance and yaw angle with respect to the webcam. Then serially communicated these parameters in a sequenced manner to Arduino UNO where these parameters are compared with the referenced input position to evaluate error signal which is then passed on to PID control mechanism. The control signal thus obtained it comprehended to vary the motor speed based on the distance coverage.

### DESIGN

The bot has been designed using Solidworks for model construction and then has been 3D printed using FDM printing.

### COMPONENTS

3D Printed Chassis, N25 Encoder Motors, Nvidia Jetson Nano, Arduino UNO, Web Camera, 7805 Voltage Regulator IC, L298 Motor Driver and Li-Ion Battery Pack. 5 volt regulated voltage is required to provide rated voltage supply to Jetson Nano, and encoder circuit of Motors, meanwhile unregulated high voltage supply is directly given to motors and motor driver

### TECH STACK

Arduino IDE, Python, OpenCV, SolidWorks

### FEATURES

- Real-Time Marker Detection: ArUco-based detection of marker ID, distance, and yaw.
- PID-Controlled Navigation: Smooth, minimal-error trajectory adjustments.
- Autonomous Docking: Accurate final docking based on marker distance.
- Live Debug Overlay: Real-time feedback on yaw, ID, and pose.

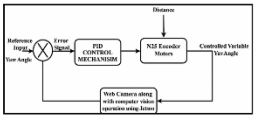
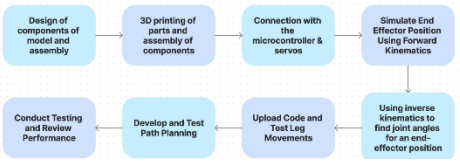



Fig. Block Diagram for closed loop control system

### METHODOLOGY




### APPLICATIONS

- Warehouse Logistics and Material Handling
- Agricultural Robots
- Defense and Security Patrols
- Space Rover Navigation
- Autonomous Delivery Systems
- Educational Robotics Kits

## Some Snapshots

